Bash survival cheat-sheet

Git, Python Environments & Coding Best Practices — MSA-DATB07-03 — works in Git Bash (Windows), Terminal (macOS) and any Linux shell

Where am I, what is here

`pwd`	print the current directory
`ls  ls -l  ls -la`	list; long format; with hidden files
`ls -lS  ls -lt`	sorted by size; by date
`cd dir  cd ..  cd ~`	go into; up; home
`cd -`	back to the previous directory
`tree -L 2`	tree view (if installed)

Paths: `/abs/olute`, `rel/ative`, `.` here, `..` parent, `~` home. Git Bash: `C:\Users` is `/c/Users`. Quote paths with spaces.

Files and directories

`mkdir -p a/b/c`	create, with parents
`touch f.txt`	create empty / update date
`cp src dst  cp -r dir dst`	copy; copy a directory
`mv old new`	move or rename
`rm f  rm -r dir`	delete (no recycle bin!)
`ln -s target link`	symbolic link

Look inside

`cat f`	print whole file
`less f`	page through (space, /search, q)
`head -n 5 f  tail -n 5 f`	first / last lines
`tail -f log`	follow a growing file (Ctrl+C)
`wc -l f  wc -w f`	count lines / words
`diff a b`	compare two files

Search

`grep word f`	lines containing word
`grep -n -i -c -v`	line numbers; ignore case; count; invert
`grep -r word .  grep -rl`	recursive; file names only
`grep -E "a|b" f`	regular expression
`find . -name "*.csv"`	files by name (wildcards)
`find . -type d  -type f`	only directories / files
`find . -size +10k -mtime -7`	bigger than; modified last 7 days

Pipes and redirection

`cmd > f  cmd » f`	write (overwrite) / append
`cmd 2> err.txt`	errors only
`cmd > out.txt 2>&1`	output and errors together
`cmd1 | cmd2`	output of 1 = input of 2
`sort  sort -n  sort -h`	sort; numeric; human sizes
`uniq -c`	collapse repeated lines, count
`cut -d, -f2  cut -c27-`	column 2 of a CSV; characters 27+
`tr a-z A-Z`	translate characters
`xargs cmd`	run cmd on each line of input
`tee f`	write to f AND pass through

Classic: `grep ERROR log | cut -c27- | sort | uniq -c | sort -nr`

Keyboard habits

`Tab`	complete names (twice: list)
`Up / Down  Ctrl+R`	history; search history
`Ctrl+C  Ctrl+L`	stop (does not copy!); clear screen
`Ctrl+Insert / Shift+Insert`	copy / paste (Windows)
`Cmd+C / Cmd+V`	copy / paste (macOS)
`drag a file onto the terminal`	types its full path
`Ctrl+A / Ctrl+E`	start / end of line
`history  !42`	list; re-run number 42
`code .  code f`	open VS Code here / on f

Environment

`echo $HOME  echo $PATH`	print a variable
`export NAME=value`	set for this shell (and children)
`env  env | grep X`	list variables
`which python`	which program will run
`alias ll='ls -la'`	shortcut (put it in ~/.bashrc)
`source ~/.bashrc`	reload config (macOS: .zshrc)

Network (Windows / macOS names)

`ping -n 4 host  ping -c 4 host`	reach it? (Windows / macOS)
`tracert -d host` `traceroute -n host`	route, hop by hop
`nslookup host`	name to IP, and the DNS used
`ipconfig /all` `ifconfig`	my addresses
`netstat -an | grep -i listen`	listening ports
`netstat -ano | findstr 8888`	Windows: PID on port 8888
`lsof -i :8888`	macOS: same
`curl -I url`	HTTP headers, status code
`curl -s url | python -m json.tool`	pretty JSON
`curl -L -o f url`	download, follow redirects
`curl -s -o /dev/null -w "%{http_code}" url`	status only
`ssh user@host  ssh -p 2222`	remote shell
`scp f user@host:/path/`	copy over ssh
`ssh-keygen -t ed25519`	create a key pair
`ssh -T git@github.com`	test the GitHub key

Status codes: 200 ok, 301/302 redirect, 401/403 not allowed, 404 not found, 5xx server error. `* * *` in a route = silent hop, not a failure. Windows error `CRYPT_E_NO_REVOCATION_CHECK`: add `--ssl-no-revoke` to curl; for Git, `git config --global http.schannelCheckRevoke false`.

Processes and disk

`ps  ps aux | grep python`	my processes; all, filtered
`top  htop`	live view (q quits)
`kill PID  kill -9 PID`	stop; force
`cmd &  jobs  fg`	background; list; foreground
`df -h`	free disk space
`du -sh * | sort -h`	size of each entry, sorted

Permissions, archives, system

`ls -l`	-rwxr-xr– owner/group/others
`chmod +x script.sh`	make executable, run with ./script.sh
`tar czf a.tar.gz dir/`	archive + compress
`tar xzf a.tar.gz  tar tzf`	extract; list
`zip -r a.zip dir/` `unzip a.zip`	zip / unzip
`whoami  hostname` `uname -a`	who, where, which OS
`date  date +%F`	now; 2026-09-18

Windows (Git Bash) vs macOS

	Windows (Git Bash)	macOS
shell check	`echo $0` = bash	`echo $0` = zsh
Python	`python`	`python3`
venv activate	`.venv/Scripts/activate`	`.venv/bin/activate`
route	`tracert`	`traceroute`
my IP	`ipconfig`	`ifconfig`
ping count	`-n 4`	`-c 4`
clipboard	`| clip`	`| pbcopy`
open folder	`explorer .`	`open .`
line endings	CRLF warning is normal	LF

Git, the daily loop (session 1)

`git init  git clone url`	new repository; copy one
`git status`	where every file stands
`git add f  git add .`	stage
`git commit -m "msg"`	snapshot the staged files
`git log --oneline --graph --all`	history
`git diff  git diff --staged`	unstaged / staged changes
`git show HEAD~1`	one commit
`git restore --staged f`	unstage (safe)
`git restore f`	discard edits (destroys)
`git revert hash`	undo with a new commit (safe)
`git reset --hard HEAD~1`	drop last commit + edits (destroys)